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SC9863A

Audio Tuning Guide

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UNISOC (SHANGHAI) TECHNOLOGIES CO., LTD.

Revision History

Version	Date	Notes
V1.0	2019/4/8	This is the initial version.
V1.1	2021/3/3	Modify slides layout and some description.

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Keyword

Keyword: Audio Tuning

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01

Overview

Overview

- This document provides basic tuning instructions of audio call parameters (including loudness rating, volume control, frequency response, noise suppression, echo cancellation and others) and multimedia (including music play, FM and audio record) parameters on Unisoc platforms.
- This document is intended for personnel who need to conduct basic audio tuning operations on Unisoc platforms. For further information about audio tuning or complex audio problems, refer to ***AudioTester Tool User Guide*** released in the toolkit or contact Unisoc audio support engineers.

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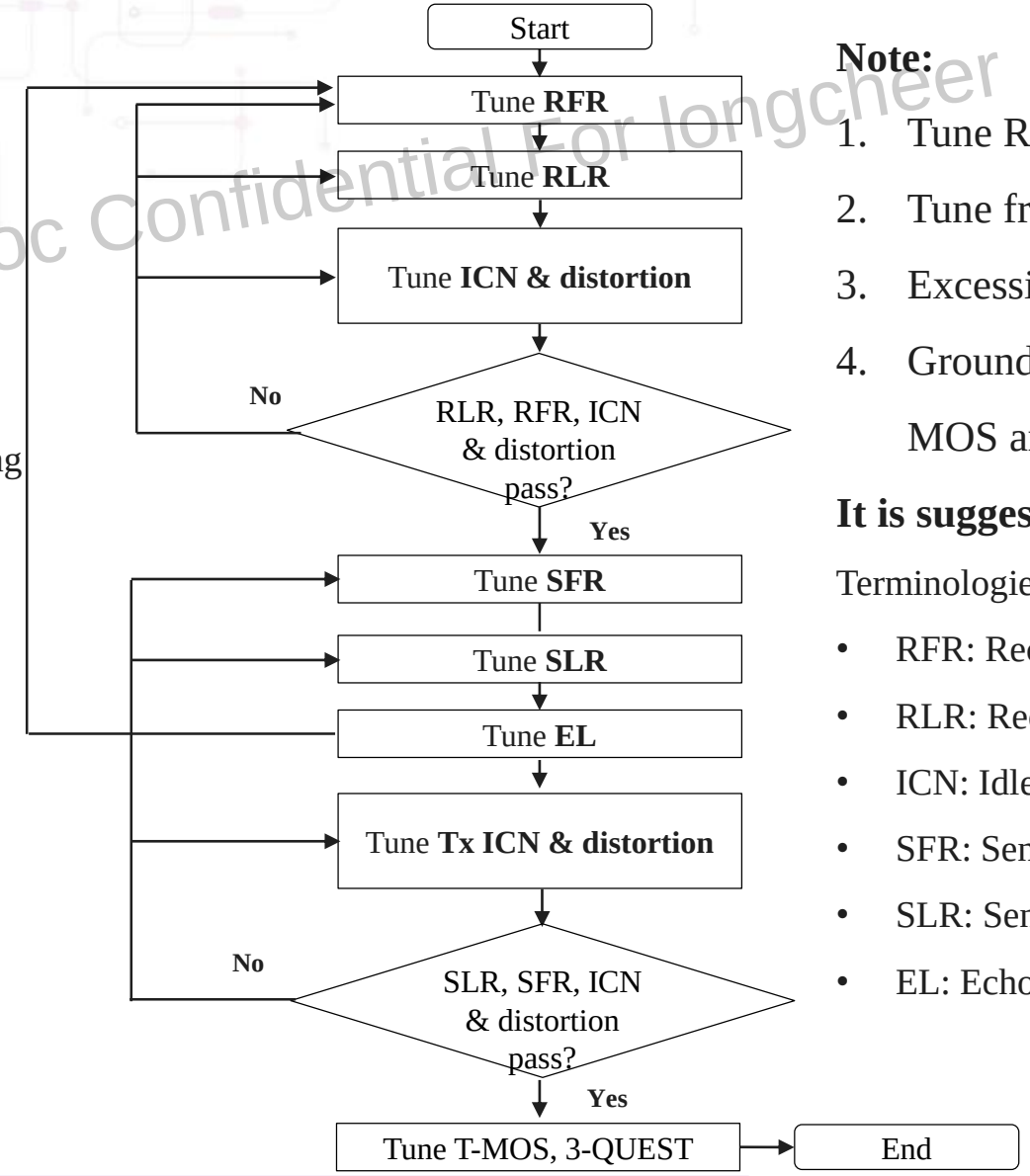
02

Audio Call Tuning

Audio Call Tuning Workflow



Echo tuning may involve optimizing RX gain value.

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Note:

1. Tune RX parameters before TX parameters.
2. Tune frequency response before loudness.
3. Excessive echo cancellation can affect TX loudness.
4. Ground noise suppression can affect the performance of T-MOS and 3-QUEST.

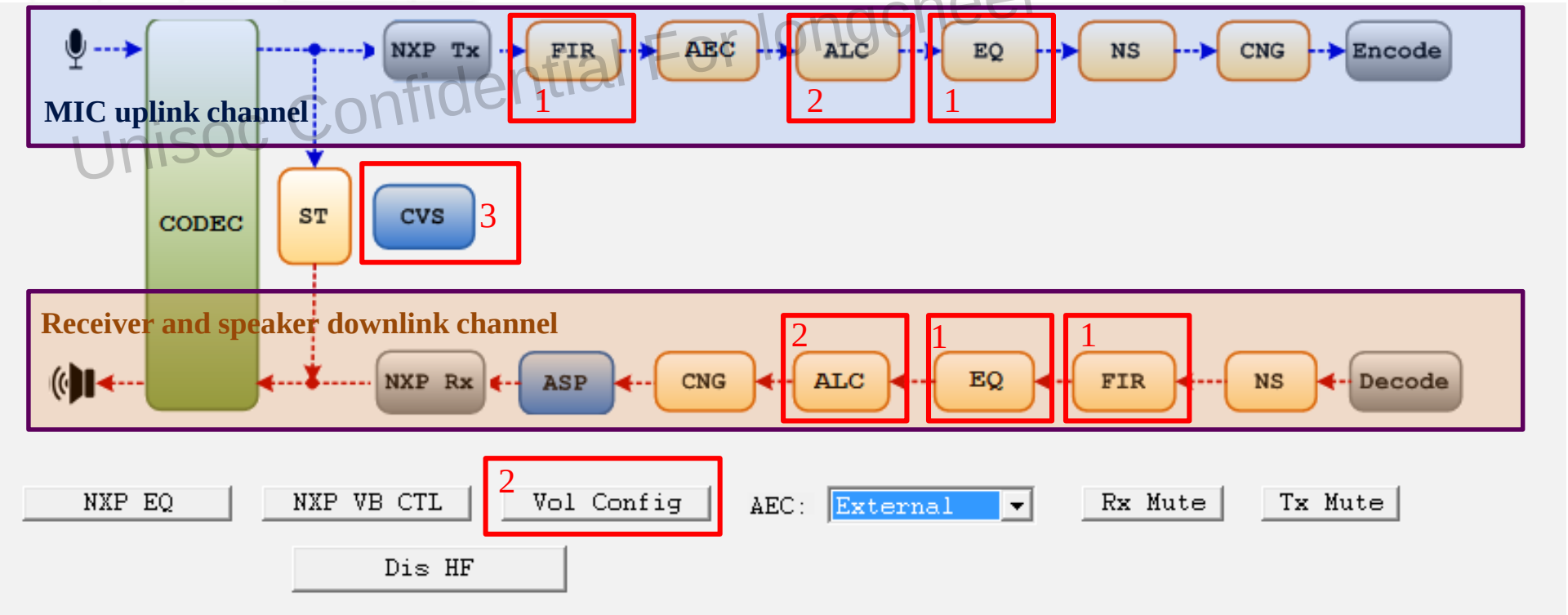
It is suggested to retest all cases after tuning all cases.

Terminologies in the workflow：

- RFR: Receiving Frequency Response
- RLR: Receiving Loudness Rating
- ICN: Idle Channel Noise
- SFR: Sending Frequency Response
- SLR: Sending Loudness Rating
- EL: Echo Loss

Audio Call Tuning – Main Interface

Main interface of audio call tuning



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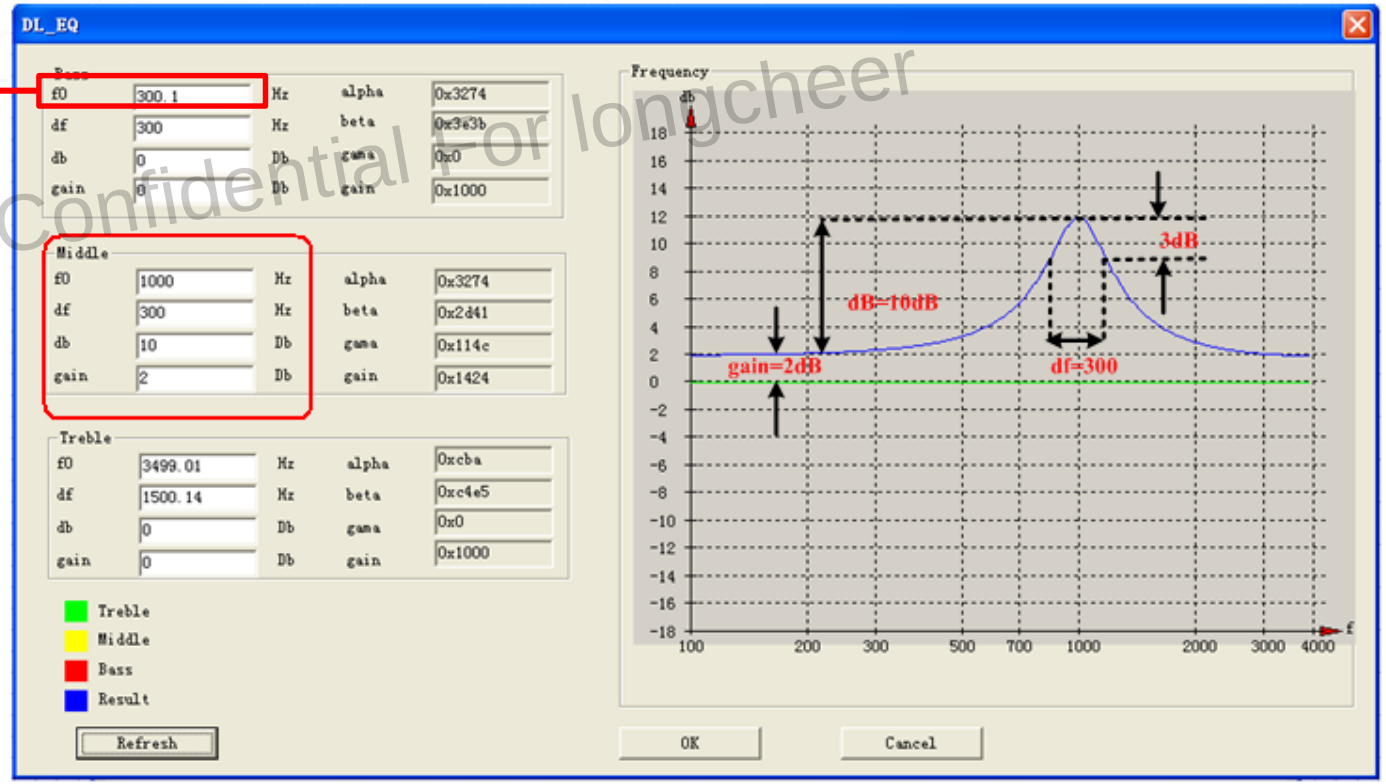
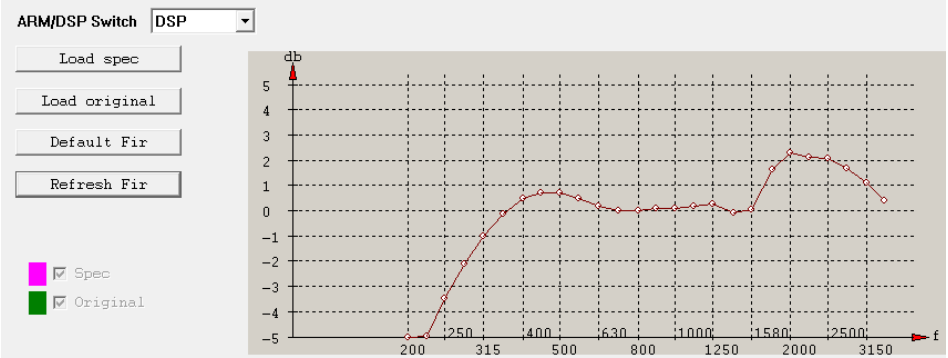
1. EQ and FIR: Frequency response tuning
2. Vol Config and ALC: Loudness tuning
3. CVS: Echo and noise suppression algorithm

Audio Call Tuning – Frequency Response

The f0 field should be no less than 50Hz, or else EQ will be abnormal.



EQ tuning interface



FIR tuning interface

Conduct flate tuning with EQ first, and then fine tuning with FIR

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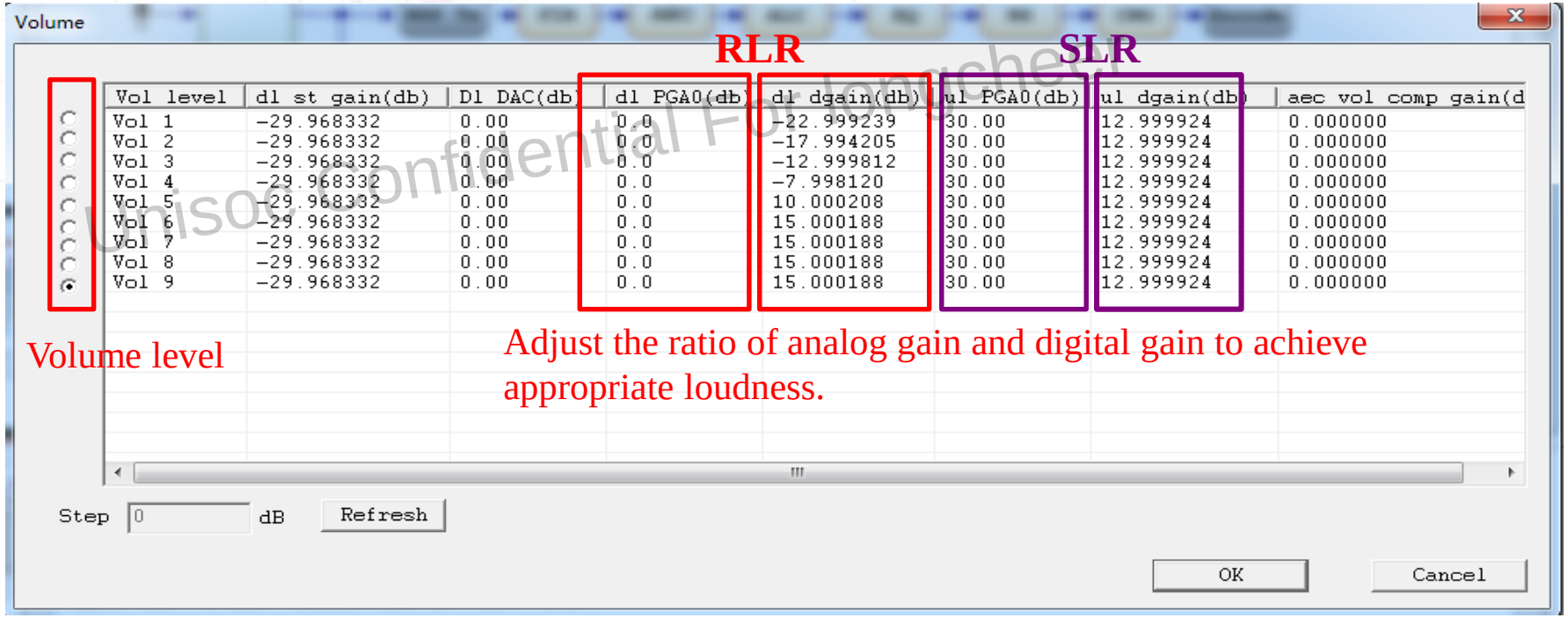
Audio Call Tuning – Loudness (1/2)



Vol level	dl st gain(db)	Dl DAC(db)	dl PGA0(db)	dl dgain(db)	ul PGA0(db)	ul dgain(db)	aec vol comp gain(d
Vol 1	-29.968332	0.00	0.0	-22.999239	30.00	12.999924	0.000000
Vol 2	-29.968332	0.00	0.0	-17.994205	30.00	12.999924	0.000000
Vol 3	-29.968332	0.00	0.0	-12.999812	30.00	12.999924	0.000000
Vol 4	-29.968332	0.00	0.0	-7.998120	30.00	12.999924	0.000000
Vol 5	-29.968332	0.00	0.0	10.000208	30.00	12.999924	0.000000
Vol 6	-29.968332	0.00	0.0	15.000188	30.00	12.999924	0.000000
Vol 7	-29.968332	0.00	0.0	15.000188	30.00	12.999924	0.000000
Vol 8	-29.968332	0.00	0.0	15.000188	30.00	12.999924	0.000000
Vol 9	-29.968332	0.00	0.0	15.000188	30.00	12.999924	0.000000

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Note:

- Downlink digital gain is protected by ALC, so it can be adjusted to above 0dB and the adjustable range is [-72,18]. Uplink digital gain is not protected by ALC, so need to consider if the value of ul_dgain is too large in case of distortion.
- It is suggested to implement downlink volume level variation in digital gain, because the variation in analog gain may cause pop noise.

Audio Call Tuning – Loudness (2/2)

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ALC compress limit and threshold can suppress large signal effectively to achieve power protection. Reduce ALC compress limit and ALC compress threshold properly in case of cracking voice or large power.

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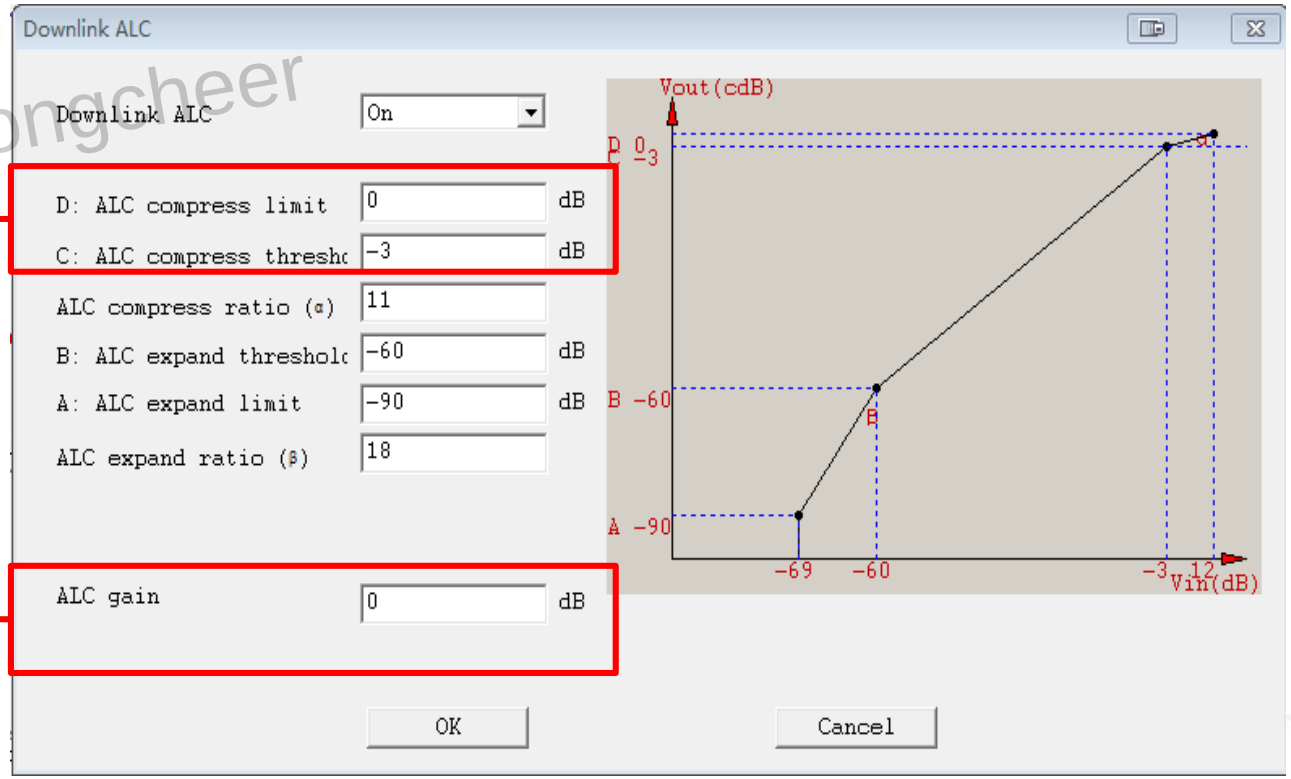
Note:

1. The value of ALC compress limit should be less than 0dB.
2. ALC compress limit is no less than ALC compress threshold.

The pre-gain of ALC
Range: -72dB ~ 18dB

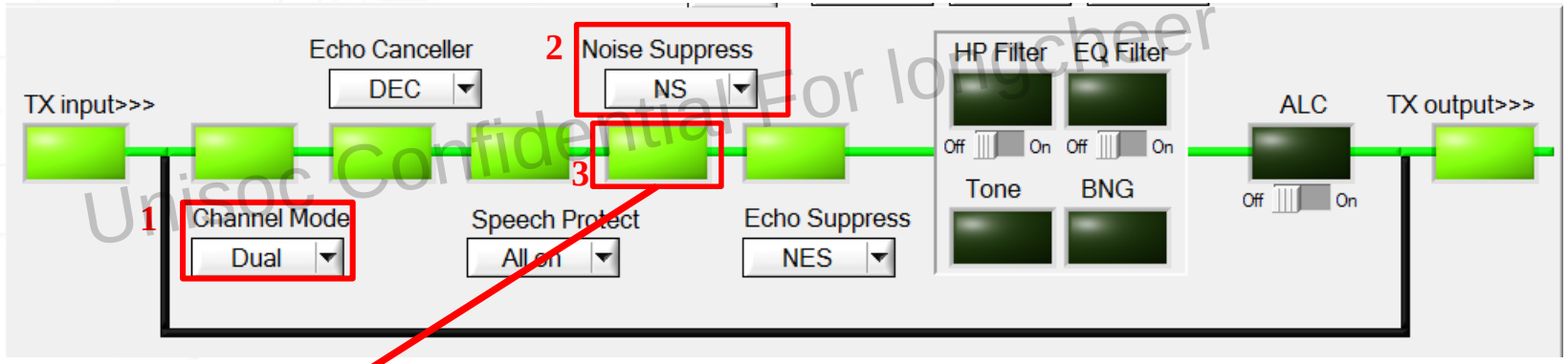


The amplitude limiter function to avoid downlink cracking voice

In addition to ALC, an amplitude limiter is designed in CVS to avoid cracking voice

- The location in NV: **AUDIO\CVS_PARAMS\Handsfree\cvs_ctrl_params\reserved\reserved[21]=0x6000**
  The blue text (like Handsfree) indicates mode, while red text (like0x6000) indicates value of the limiter.
- The way to set limiter value: The value in NV is hexadecimal, with the maximum value 0x7fff corresponding to 0dB. Half of 0x7fff corresponds to -6dB, a quarter of 0x7fff corresponds to -12dB, and so on in a similar fashion.

Audio Call Tuning – Ground Noise



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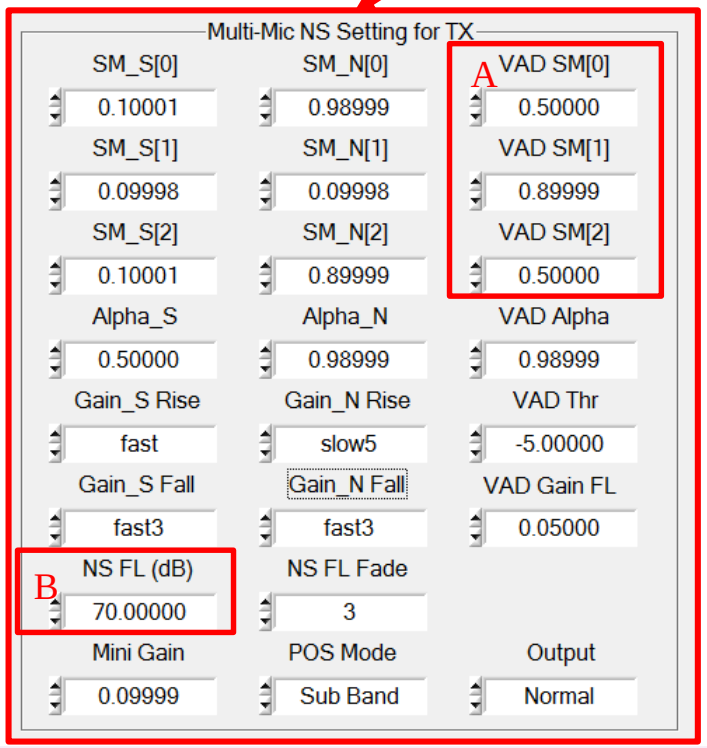
1. Select channel mode.
   - Dual: dual mic
   - Mono: single mic
2. Select noise suppression mode
   - Bypass: Noise suppression off
   - NS: Normal noise suppression
   - NS+: Enhanced noise suppression



3. Tune multi-mic NS setting for TX, which mainly involves parameters in rectangle A and B.
   A.VAD_SM[0]/VAD_SM[1]/VAD_SM[2]: Judge audio frame and noise frame
   - The greater the value of VAD_SM[0]/VAD_SM[2], the more possibility to be judged as noise; Recommendation: 0.1 ~ 0.5; Step: 0.1.
   - The smaller the value of VAD_SM[1], the more possibility to be judged as noise; Recommendation: 0.9 ~ 0.95; Step: 0.01.

   B.NS FL(dB): Cancel background noise less than this value
   - The greater the value, the greater the noise suppression.
   - Range: 0 ~ 128; Recommendation: 70 ~ 100; Step: 1.

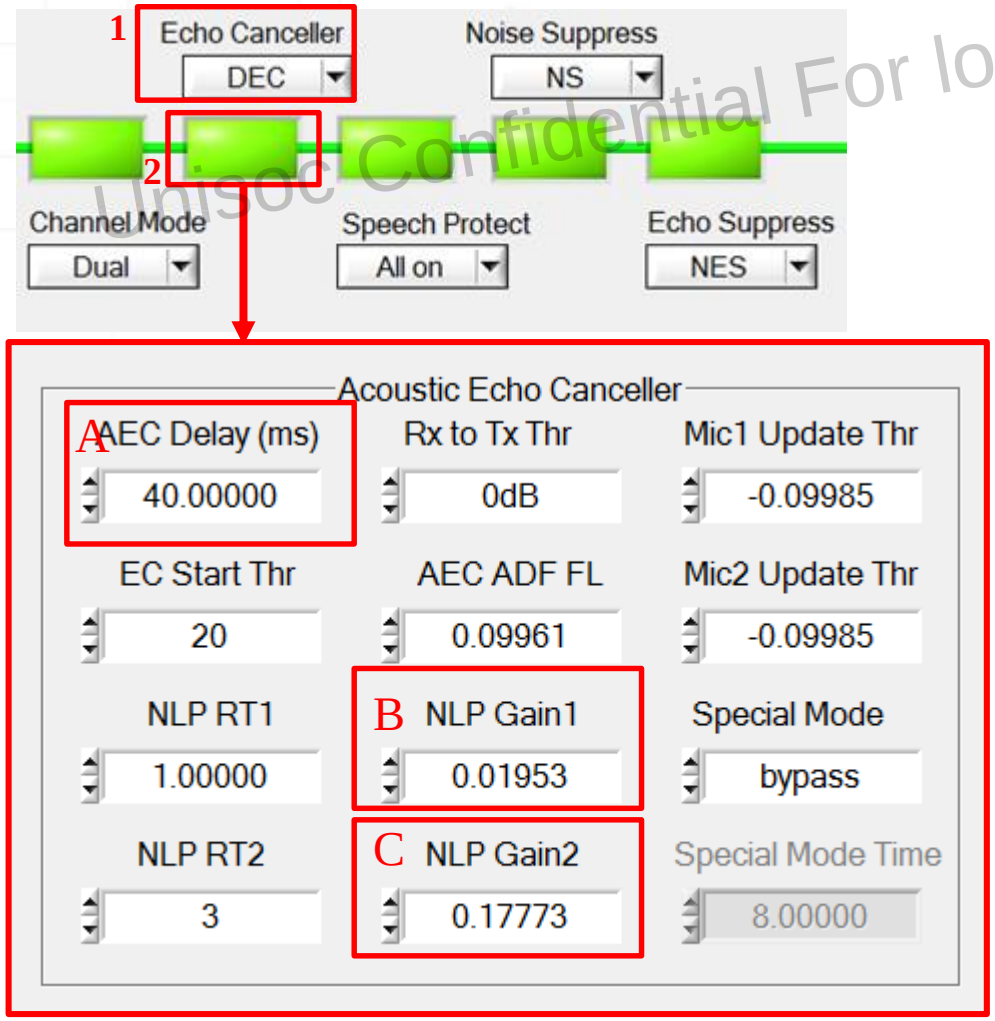
Uplink TDNC Algorithm:

The filter is added in DSP to suppress uplink TDD noise, and the location in NV is as follows:

audio\audio\Handset\AudioStructure\extend\extend[83]=0x1000

- Blue text (like Handset) indicates mode.
- Red text indicates filter switch status, with Bit12=1 for on and Bit12=0 for off.

Audio Call Tuning – Linear Echo



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1. Select an appropriate filter mode, usually four modes, namely DEC, DEC+, DECs and DECs+
2. Set the parameters of the selected filter mode, mainly the parameters in rectangle A, B and C.

A.AEC Delay: The time gap between downlink reference signal and uplink signal. The empirical value is 43.5ms, which can be fine tuned properly.

B.NLP Gain1: The ratio between uplink echo signal and downlink reference signal of main mic.

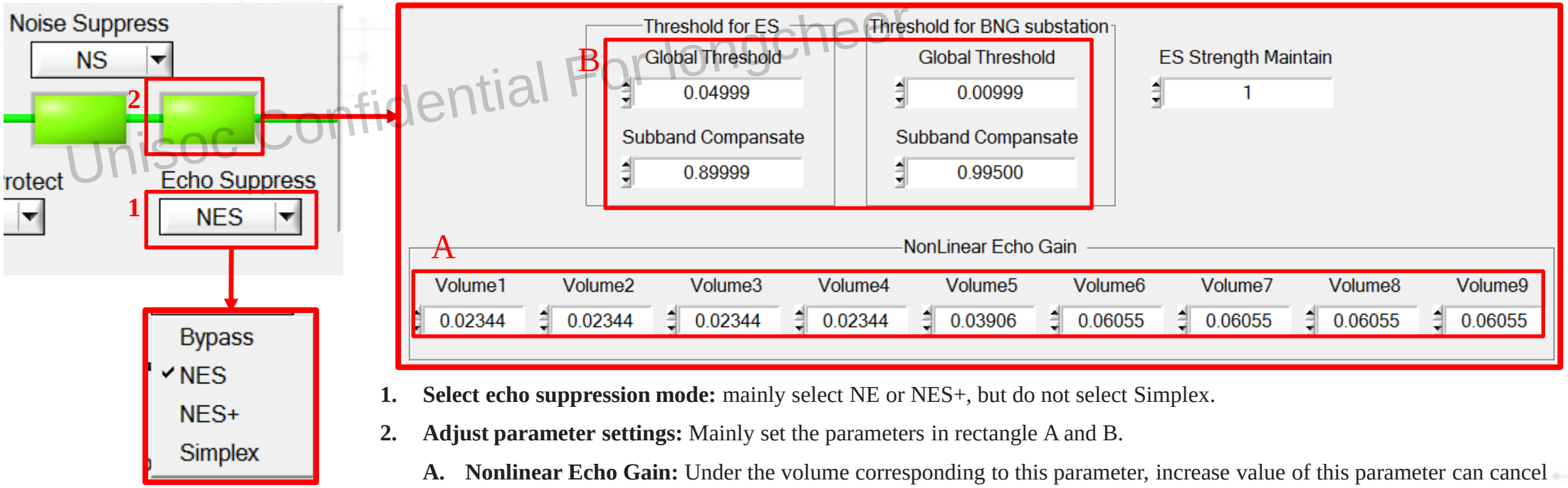
- Handset mode: usually set to 0.01 ~ 0.05.
- Handsfree mode: usually set to 0.1 ~ 0.8. For external power amplifier solution, the value can be larger.

C.NLP Gain2: This parameter has different meanings for mono mic and dual mic solutions.

✓ **Mono: The greater the value, the greater the echo suppression.** Handset: usually set to 0.05 ~ 0.2; Handsfree: usually set to 0.3 ~ 1.5.

✓ **Dual: The ratio between uplink echo signal and downlink reference signal of auxiliary mic** . Only handset mode, usually set to 0.1 ~ 0.3.

Audio Call Tuning – Nonlinear Echo



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1. **Select echo suppression mode:** mainly select NE or NES+, but do not select Simplex.
2. **Adjust parameter settings:** Mainly set the parameters in rectangle A and B.

   A. **Nonlinear Echo Gain:** Under the volume corresponding to this parameter, increase value of this parameter can cancel echo effectively. Range**:** 0 ~ 64; Recommendation: 0.05~0.1 for handset, 0.1 ~ 0.5 for handsfree, and appropriately increase the value for external power amplifier.

   B. **Global threshold for ES** is the suppression threshold of sub-band echo. The greater the value, the more possibility to be in sub-band echo; Range**:**0 ~ 1; recommendation: 0.05 ~ 0.1 for handset, 0.1 ~ 0.3for handsfree.

   Global threshold for BNG is the residual threshold after sub-band echo suppression. If the residual is below this value, fill comfortable noise. The greater the value, the more comfortable noise filled. Recommendation**:** 0.01 ~ 0.03 for handset, and 0.05 ~ 0.1 for handsfree.

   The threshold for ES should be greater than the threshold for BNG.

   Subband compensate is for subband compensation. The greater the value, the less the compensation and the less the echo suppression; Recommendation: 0.8 ~ 0.99.

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03

Multimedia Tuning

Multimedia Tuning

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There are 3 tuning modes on multimedia tuning interface:

- Music Play
- FM
- Audio Record

Select corresponding mode before multimedia tuning.

Music Play Tuning – Loudness

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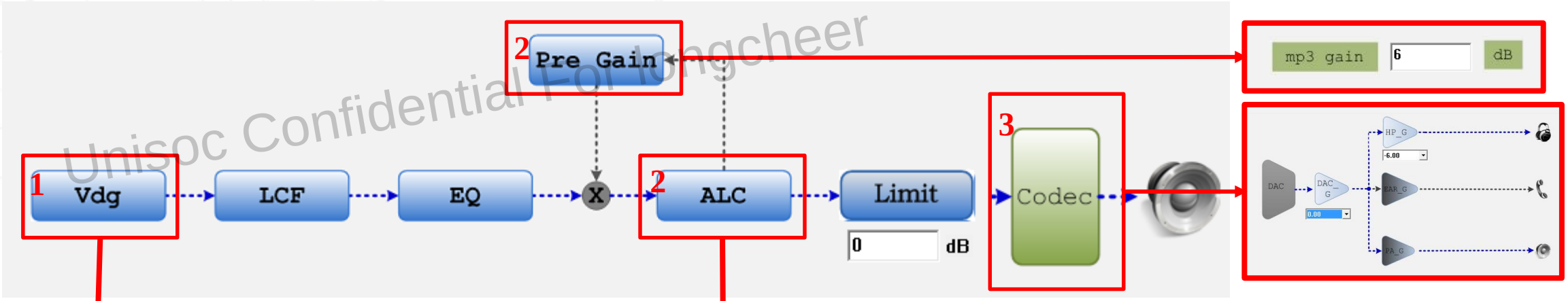
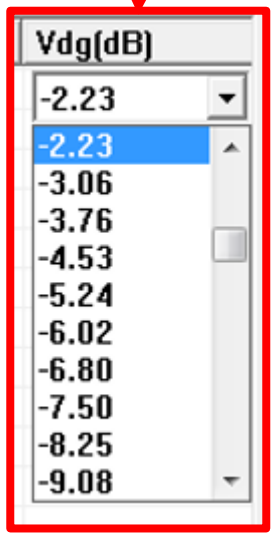
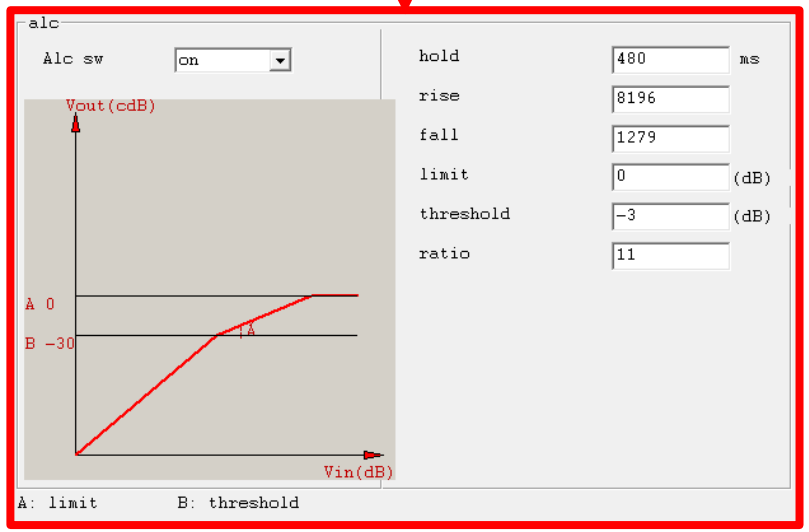






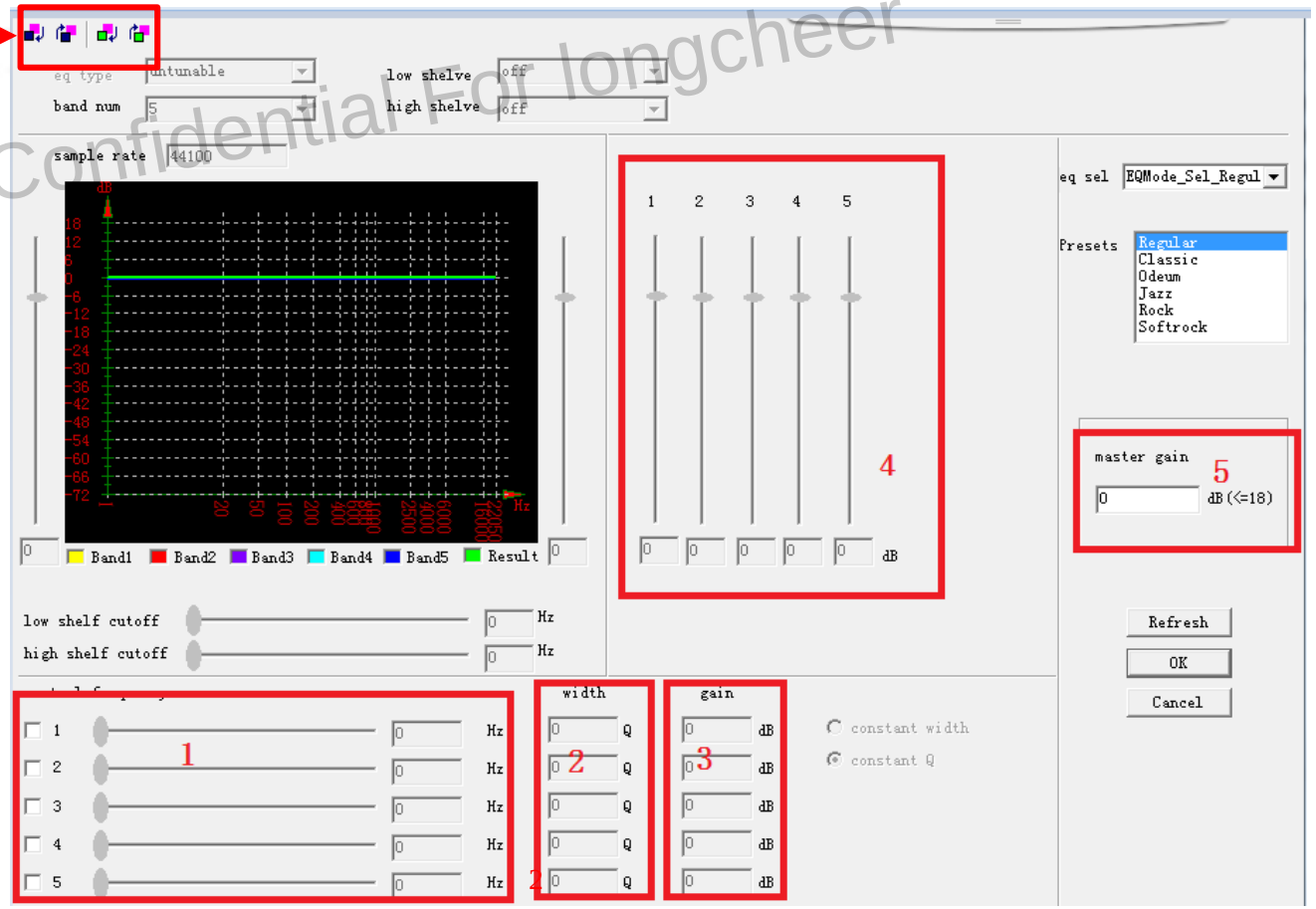
Steps to tune music playing parameters:

1. Set the value of Vdg. The value is not protected by ALC. Suggest to set it to below 0dB. Recommendation: -0.78.
2. Set the value of Pre Gain and ALC. Amplify small signals, and protect large signals from clipping distortion. The recommended value of Limit is no greater than 0dB.
3. Set an appropriate Codec analog gain.

Music Play Tuning – EQ

Writing of EQ Parameter:

The writing of EQ is independent. Each curve updating requires the writing operation on EQ interface.



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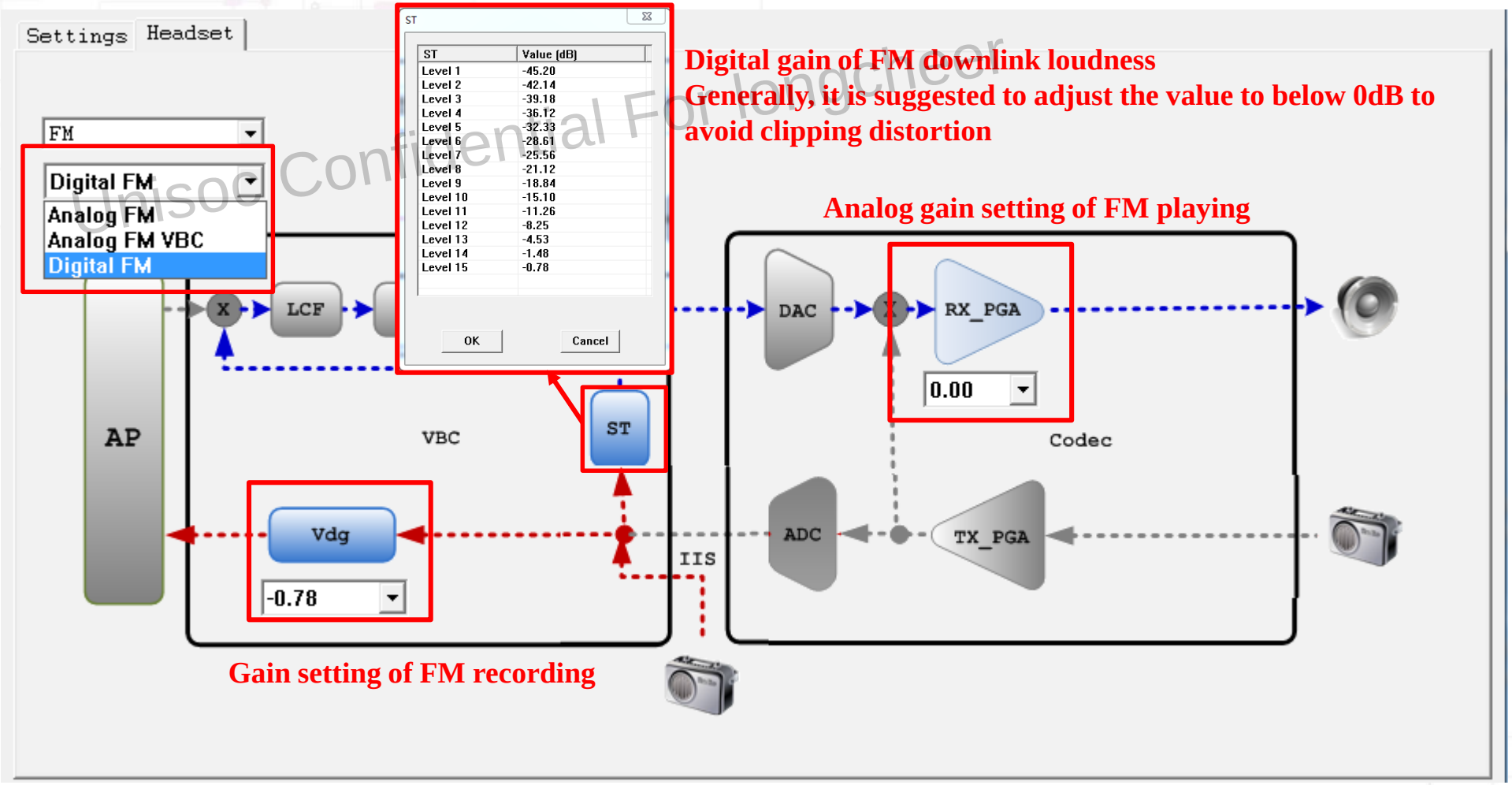
EQ Parameters:

1. Central frequency f0
2. Band width setting factor Q of equalizer
3. Overall gain of equalizer
4. Central frequency gain of equalizer
5. Master gain

Tips:

EQ is not protected by ALC. It is suggested to set master gain to be a negative value first, and then adjust EQ to prevent unobvious EQ tuning effect or distortion caused by saturation.

FM Tuning – Loudness



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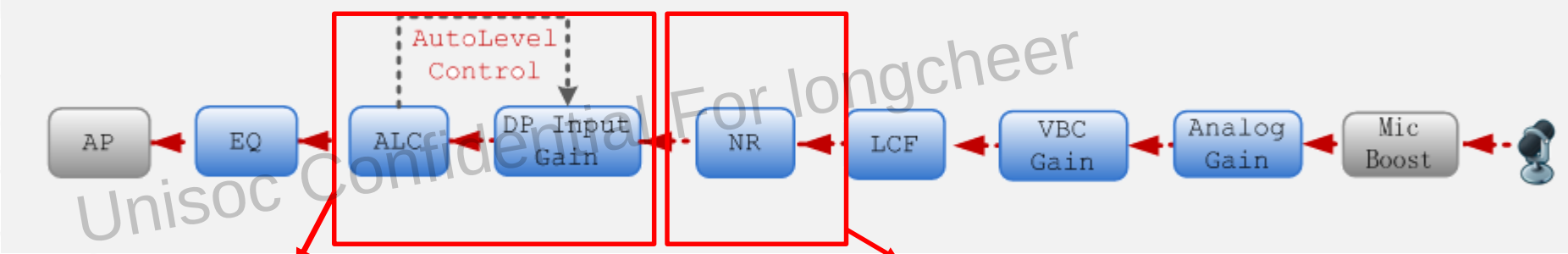
Note:

- Smartphones use digital FM currently.
- FM and music play shares the same sound effect parameters (LCF, EQ, ALC, Limit).

Audio Record Tuning – Ground Noise



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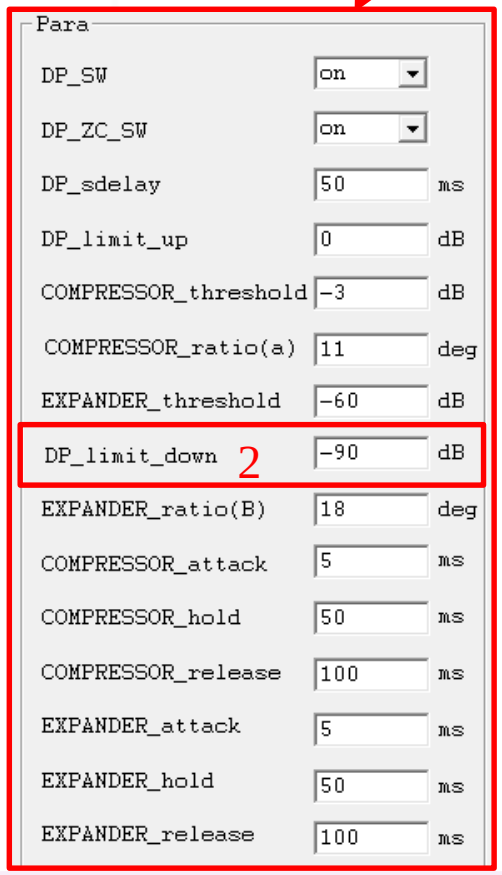
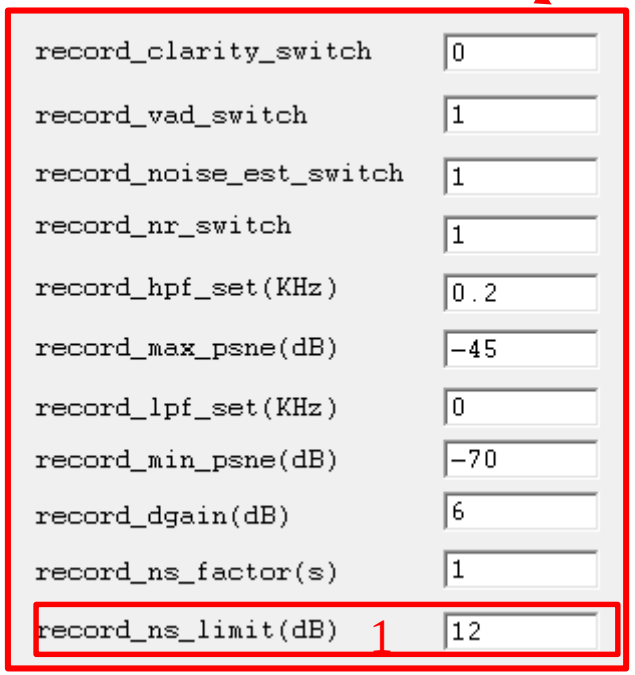


Para		
DP_SW	on	
DP_ZC_SW	on	
DP_sdelay	50	ms
DP_limit_up	0	dB
COMPRESSOR_threshold	-3	dB
COMPRESSOR_ratio(a)	11	deg
EXPANDER_threshold	-60	dB
DP_limit_down 2	-90	dB
EXPANDER_ratio(B)	18	deg
COMPRESSOR_attack	5	ms
COMPRESSOR_hold	50	ms
COMPRESSOR_release	100	ms
EXPANDER_attack	5	ms
EXPANDER_hold	50	ms
EXPANDER_release	100	ms



record_clarity_switch	0
record_vad_switch	1
record_noise_est_switch	1
record_nr_switch	1
record_hpf_set(KHz)	0.2
record_max_psne(dB)	-45
record_lpf_set(KHz)	0
record_min_psne(dB)	-70
record_dgain(dB)	6
record_ns_factor(s)	1
record_ns_limit(dB) 1	12

Suggest to tune the following parameters in case of great ground noise in audio recording.

1. record_ns_limit in NR: It is recommended to be 0 ~ 12.
2. DP_limit_down: Algorithm cancels the ground noise below this value. The value of this parameter should be smaller than that of **EXPANDER_threshold.**

Tips:

Too large analog gain may cause greater ground noise. To solve the issue, try to reduce analog again appropriately and increase digital gain. However, excessive reduction of analog gain and increase of digital gain may reduce audio quality, so the adjustment should be appropriate.

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